PART II: PATTERNS

Topic 20: How to Read a Time Series Chart — A Checklist

The Core Concept

This final topic pulls together everything from Parts I and II into a practical, systematic checklist. When you pick up any time series chart — whether in a briefing, a report, or a news article — this is the order in which to read it.

The 12-Question Checklist

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

#	Question to Ask	What You Are Looking For
1	**What is the time unit?**	Hourly / daily / monthly / yearly? Match to your question.
2	**Is the Y-axis starting at zero?**	If not, visual changes may be exaggerated.
3	**Is there a trend?**	Long-term direction: up, down, or flat?
4	**Is there seasonality?**	Do the same peaks/troughs repeat each year?
5	**Are there outliers?**	Any point far outside the normal range?
6	**Is there missing data?**	Any gaps in the line? Investigate why.
7	**Are there multiple series?**	Do they move together (external cause) or apart (local cause)?
8	**What is the count vs. the rate?**	Does activity volume explain the change?
9	**What does last year look like?**	YoY comparison removes seasonal effects.
10	**Could there be a lag?**	Did a policy change precede this shift by 3–6 months?
11	**Is this trend or noise?**	Would a 3-month moving average change my conclusion?
12	**Could data quality explain this?**	System changes, definition shifts, reporting lags?

Putting It All Together

Good analysis does not require math. It requires asking the right questions in the right order. Every concept in this checklist corresponds to a topic in this tutorial series:

- Questions 1–2: Topic 3 (Visualization) and Topic 4 (Frequency)
- Questions 3–5: Topics 7, 8, 10, 11 (Trend, Seasonality, Noise, Outliers)
- Questions 6–7: Topics 6, 14 (Missing Data, Multiple Series)
- Questions 8–9: Topics 5, 18 (Rates, Year-over-Year)
- Questions 10–11: Topics 15, 13 (Lag, Moving Averages)
- Question 12: Topic 19 (Data Quality)

💡 **The Insight:** A structured reading habit turns raw charts into actionable intelligence. This checklist is your starting point — with practice, it becomes instinct.

Key Takeaway: Print this checklist. Keep it with your reporting toolkit. Use it every time you encounter a new time series chart.